2025/ 01 / 11

X

Preliminary Examination - 2025

Science and Technology (Part -II) (72)

Time - 2 hrs. **pages - 05** **Marks : 40**

Note : -
i) **All questions are compulsory.**
ii) **Use of calculator is not allowed.**
iii) **The numbers to the right of the questions indicate full marks.**
iv) **In case of MCQs (Q.No.1(A) only the first attempt will be evaluated and will be given credit.**
v) **For every MCQ, the correct alternative (A),(B),(C) or (D) with subquestion number is to be written as an answer.**
for Eg- i) (A), ii) (B) , iii) (C)
vi) **Scientifically correct,labelled diagrams should be drawn wherever necessary.**

Q.1 A) Choose the correct alternatives and write its alphabet for each of the following questions. **05**

i) The proof for the fact that protein synthesis occurs through..... was given by George Beadle and Edward Tatum.
A) Mutation B) RNA
C) Gene D) transcription

ii) Length of sperm is about .....micrometers.
A) 60 B) 6.5
C) 6 D) 7.5

iii) The apex consumer is .....
A) Eagle B) frog
C) Snake D) Gross hopper.

iv) Animal with asymmetrical body is .....
A) insects B) starfish
C) birds D) amoeba

v) ..... is example of rare species.
A) Lion tailed monkey B) Shekru
C) Tiger D) Musk deer.

B) Answer the following questions as per the given instructions. **05**

i) State whether the following statement is true or false.
Meditation helps the students to improve concentration in their studies.

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B) ii) Write the correct co-relation :
White revolution : Dairy : : Dwarfness : .....

iii) To which class does Petromyzon belong?

iv) Match the correct pair :

Column 'A'	Column 'B'
a) Western Ghat	a) Rhino b) Asiatic lion c) deforestation

V) Identify the embryos.



1 2

Q2. A) Give Scientific reasons.(any two) 04

i) Cell division is one of the important properties of cell and organisms.

ii) It is absolutely necessary to control the fission reaction in nuclear power plants.

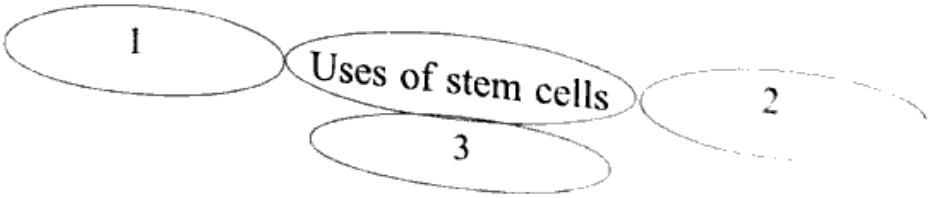
iii) Some women can take the help of surrogate mother.

B) Answer the following questions. (Any three) 06

i) Write a short note on - Natural and Artificial envirnonment

ii) Distinguish between
butterfly and bat with the help of four properties.

iii) Write the correct answer in blank circles.



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B) iv) Write any two points about pre - disaster management.

v) What are the benifits of Xanthan gum?

Q.3) Answer the following questions. (any five) **15**

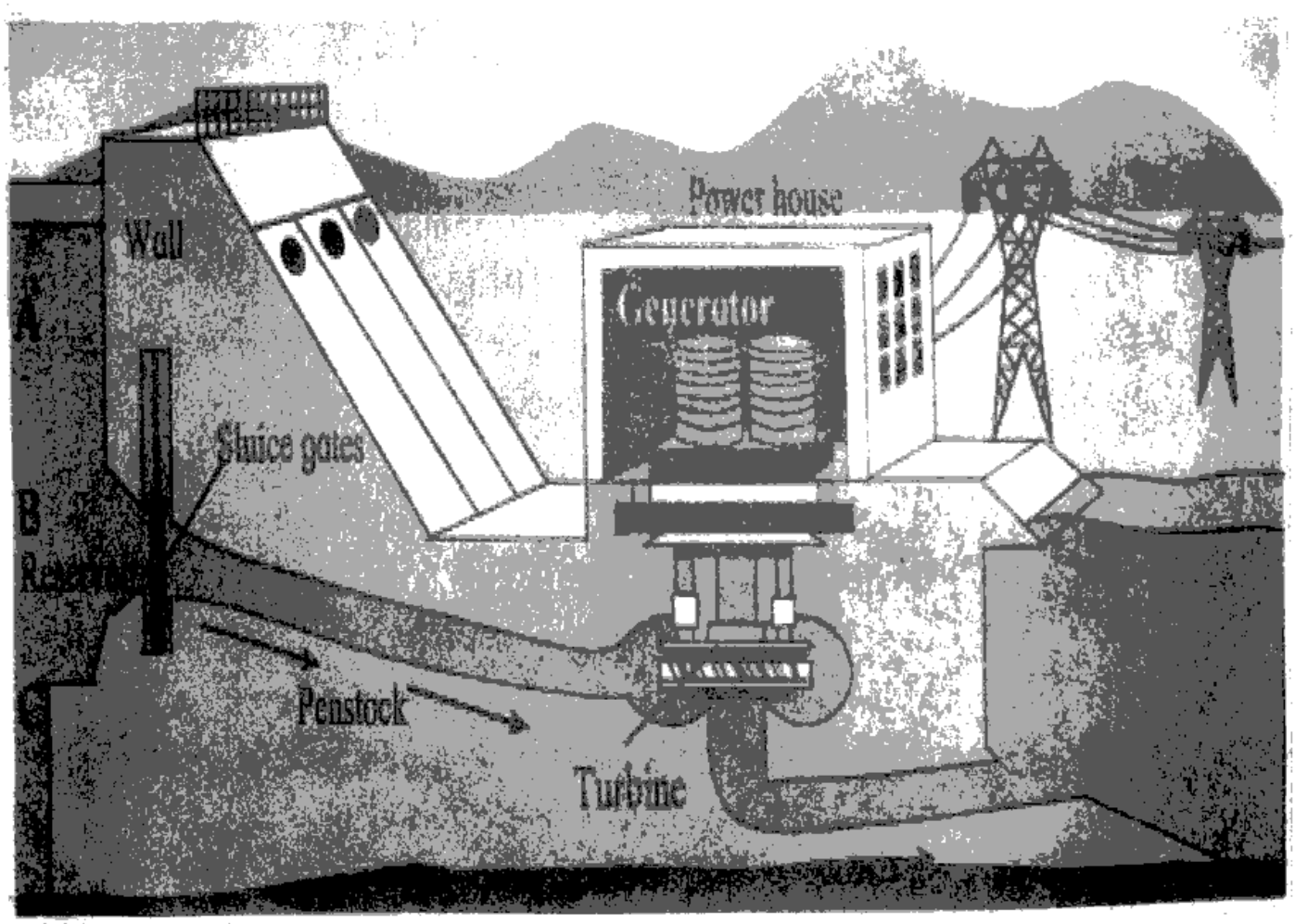
i) Explain the theory of evolution and mention the proof supporting it.

ii) Complete the following chart.

Sr. No	steps of karyokinesis	status of chromosomes
1)	Prophase	.....
2)	.....	complete their condensation
3)	Anaphase	.....

iii) Write source and effects of Water pollution.

iv) Answer the following questions from figure.



a) With reference to point B,potential energy of how much water reservoir in the dam will be converted into kinetic energy?

b) What will be the effect on electricity generation,if the channel taking water to turbine starts at point A?

c) What will be the effect on electricity generation, if the channel taking water to turbine starts at point C?

v) Write the characters of the 'penguin ' with the help of classification chart.

vi) Identify the organs of human body that can be donated.



1 2 3 4 5

vii) How can the soil polluted by acid rain be made fertile again?

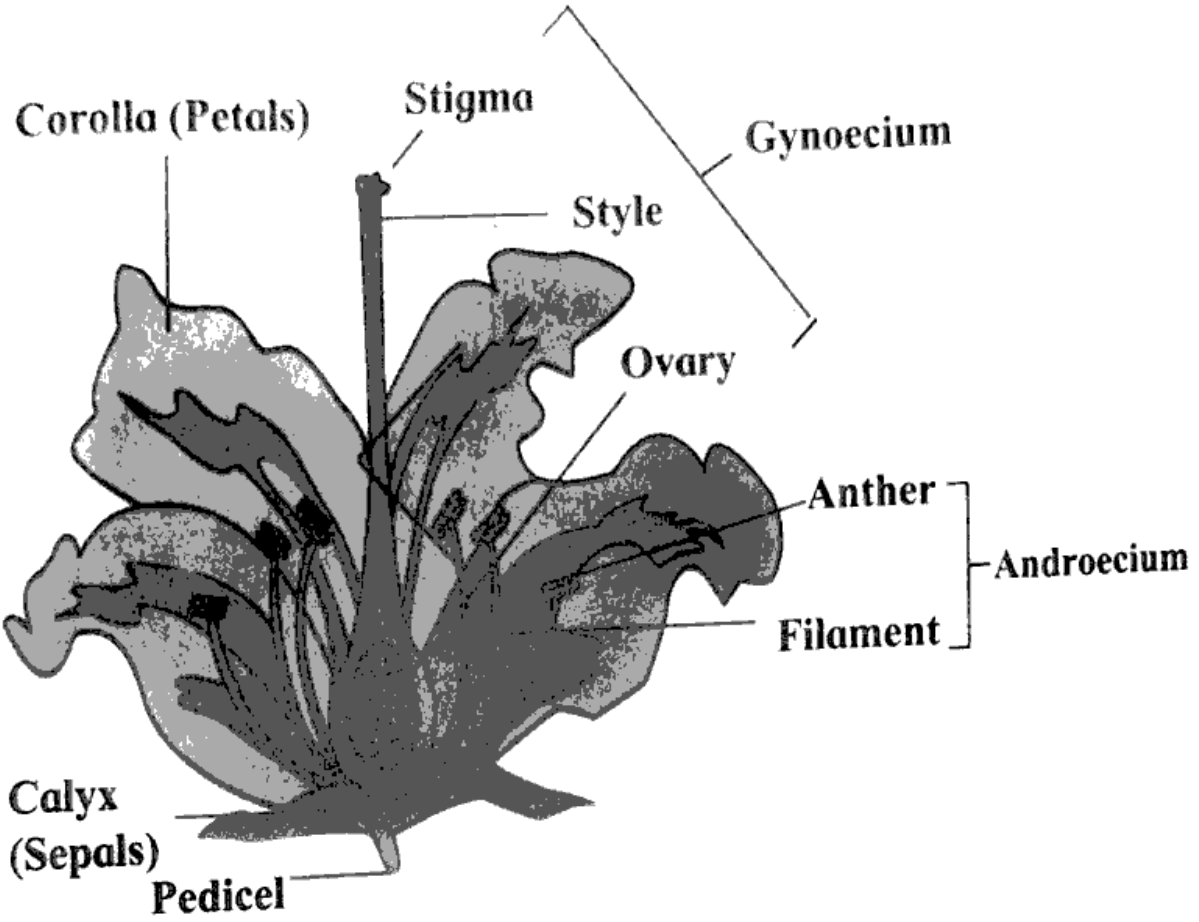
viii) Complete the following passage by using appropriate words given in brackets : (Delhi, 12,2005,army ,National Disaster, rescue work, state reserve police , 2015)

National Disaster Response Force has been established as per the Disaster Management Act, ... Divisions of this force are working in army.Overall ....divisions are working in the country. Its headquarter is in .....and it is in action all over the country with the help of .....in Maharashtra,.....Response force is in action through state reserve police force.Personnel of this force have substantial contribution in ..... in disasters like cyclones,cliff - sliding ,building collapse,etc

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Q.4) Answer any one of the following questions: **05**

i) Answer the following questions by observing the diagram.



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a) How many whorls are there in the flower from outside to inside ? 02

b) Write the name of each whorl. 01

c) Name the components of Gynoecium. 01

d) Members of carolla are known as ? 01

ii) What will you do? Why?

a) Child of your neighbour is addicted to tobacco chewing. 01

b) Your sister has become in communicative.She prefers to remain alone. 02

c) Your brother studying in XII has developed the stress. 01

d) Your friend has developed the hobby of snapping selfies. 01

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